

carbon isotopes ^{13}C and ^{12}C (expressed as $\delta^{13}\text{C}$ values; -80 to -50‰) [8,9] and MOB are known to discriminate against the heavier ^{13}C when metabolizing CH_4 , resulting in even lower $\delta^{13}\text{C}$ values of MOB biomass [10]. These very low values do not occur in aquatic and terrestrial photosynthetic primary producers (-35 to -10‰) [11–15]. The very low observed $\delta^{13}\text{C}$ values of, for example, larvae of non-biting midges (Chironomidae) of the tribe Chironomini [16,17] and planktonic water fleas of the genus *Daphnia* (Cladocera) [18] in some lakes are therefore considered a clear indication of MOB, or organisms feeding on MOB, forming a relevant part of the diet of these organisms.

Planktonic filterers such as *Daphnia* can graze MOB from the water column during stratification, a process which can effectively reduce MOB biomass and lead to increased epilimnetic CH_4 concentrations at least in some shallow boreal lakes [19]. Furthermore, *Daphnia* has been shown to rely strongly on MOB-derived carbon during autumn overturning, when the CH_4 stored in deep anoxic water layers of stratified lakes comes into contact with oxygen [20]. Benthic invertebrates that can incorporate CH_4 -derived carbon, such as chironomid larvae of the tribe Chironomini, either feed on MOB in the sediments (deposit feeders) or MOB associated with suspended organic particles (filter feeders). Some Chironomini larvae have been shown to actively maintain an oxic–anoxic interface within their tubes, providing a habitat for MOB which they feed on [21]. $\delta^{13}\text{C}$ values of chironomids and *Daphnia* closely reflect those of their food source (differences of 0 – 1‰) and of their fossilizing chitinous structures (reported offsets of 0 – 1‰) [22–25]. Chitinous remains deposited and buried in the lake sediments retain their original isotopic composition [23]. Hence, analysis of $\delta^{13}\text{C}$ values of ‘fossil’ chironomid and *Daphnia* remains can provide insights on their past food sources and into whether CH_4 -derived carbon formed a major component of their diet [23,26–31]. Available studies indicate that chitinous remains with low $\delta^{13}\text{C}$ values are deposited in lakes with high surface and deep water CH_4 concentrations and diffusive CH_4 emissions [28,29].

It is unclear whether an increase in nutrient loading, in addition to a higher CH_4 output, also results in a higher contribution of CH_4 -derived carbon to the lake food webs. Higher photosynthetic primary productivity associated with higher nutrient loading can increase the availability of algal organic matter. As a consequence, a higher proportion of algae in the diet of invertebrate groups that may also incorporate CH_4 -derived carbon could be expected. However, higher algal productivity can also lead to decreased oxygen concentrations in lake sediments and deep water layers, and associated increases in CH_4 production and CH_4 availability in lake ecosystems. This can favour the growth of MOB and their temporal and spatial availability within lakes. Therefore, how the relevance of CH_4 -derived carbon in the food web of lakes changes under influence of (past or future) changes in nutrient concentrations and productivity remains poorly constrained, particularly on decadal time scales which are not covered by instrumental measurements of CH_4 concentrations and $\delta^{13}\text{C}$ values in aquatic ecosystems.

The $\delta^{13}\text{C}$ values of chitinous remains of aquatic invertebrates were previously mainly studied in oligo- to mesotrophic, often remote lakes [27,29,30,32]. Here, we present a study of the $\delta^{13}\text{C}$ values of fossil Chironomini head capsules and *Daphnia* resting eggs (ephippia) from recent

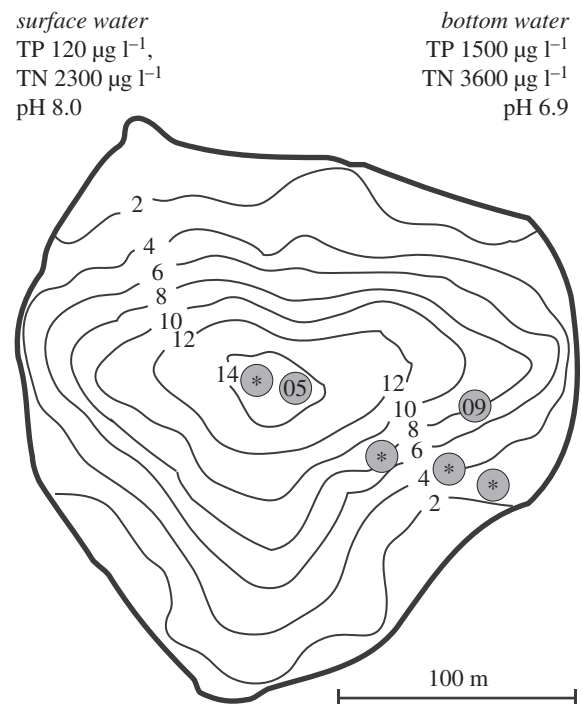


Figure 1. Bathymetric map of Lake De Waay. The circles indicate the coring locations (see the electronic supplementary material): (09) core WAY09 presented in this study, (05) core WAY05 investigated by Kirilova *et al.* [33] from which we used the XRF data (see below). Asterisks (*) indicate locations of the surface sediment samples at 2.5, 5, 8 and 14.5 m depth. Lake water nutrient concentrations and pH were measured in August 2011. TP, total phosphorus; TN, total nitrogen.

(twentieth century) sediments from a small and presently hypertrophic dimictic lake in The Netherlands (figure 1). The study lake, Lake De Waay, underwent a transition from eutrophic (total phosphorus (TP) $\sim 100\ \mu\text{g l}^{-1}$) to more mesotrophic conditions (TP $\sim 40\ \mu\text{g l}^{-1}$) and then again to hypertrophic conditions (TP $> 100\ \mu\text{g l}^{-1}$) during the past approximately 100 years [33,34]. The impact of twentieth-century temperature changes on lake ecosystems in The Netherlands is likely to have been relatively minor compared with direct anthropogenic environmental disturbances. Therefore, this study provides an opportunity to investigate the effects of both oligotrophication and (re-)eutrophication and the associated changes in oxygenation regime on the contribution of CH_4 -derived carbon to the aquatic food web, and indirectly on in-lake dissolved CH_4 availability, under relatively stable climatic conditions. We compare fossil invertebrate $\delta^{13}\text{C}$ values with a diatom-inferred reconstruction of total phosphorus concentrations (DI-TP) in the lake water and the Fe:Mn ratio of the sediments, which is expected to increase with decreasing oxygen availability at the sediment–water interface [35]. If CH_4 -derived carbon became more relevant for Lake De Waay’s food web under conditions with higher nutrient availability, we expect to see a positive relationship between nutrient availability (DI-TP) and anoxia (Fe:Mn ratio), which in turn are expected to be negatively related to $\delta^{13}\text{C}$ values of the examined invertebrate groups that can incorporate CH_4 -derived carbon. Conversely, if higher availability of algal material in the eutrophic to hypertrophic phases led to a lower relevance of CH_4 -derived carbon for the investigated aquatic invertebrates, we expect to see positive relationships between the $\delta^{13}\text{C}$ values of the studied invertebrate groups and DI-TP and Fe:Mn in Lake De Waay.